

DESK RESEARCH - MAP VISUALIZATION FOR AGING MAPS

Date: February 2026

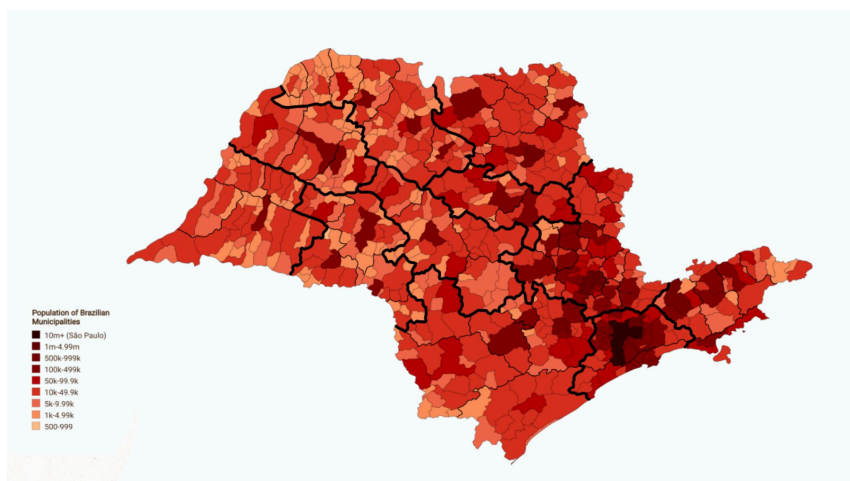
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Introduction

The following document synthesizes evidence from cartography, geovisualisation research and accessibility guidelines into concrete design decisions for the Aging Maps platform. It covers appropriate map types, techniques for representing uncertainty, the modifiable areal unit problem, accessibility for older users and practical recommendations for data governance and user interface design.

1. Choosing and designing map types

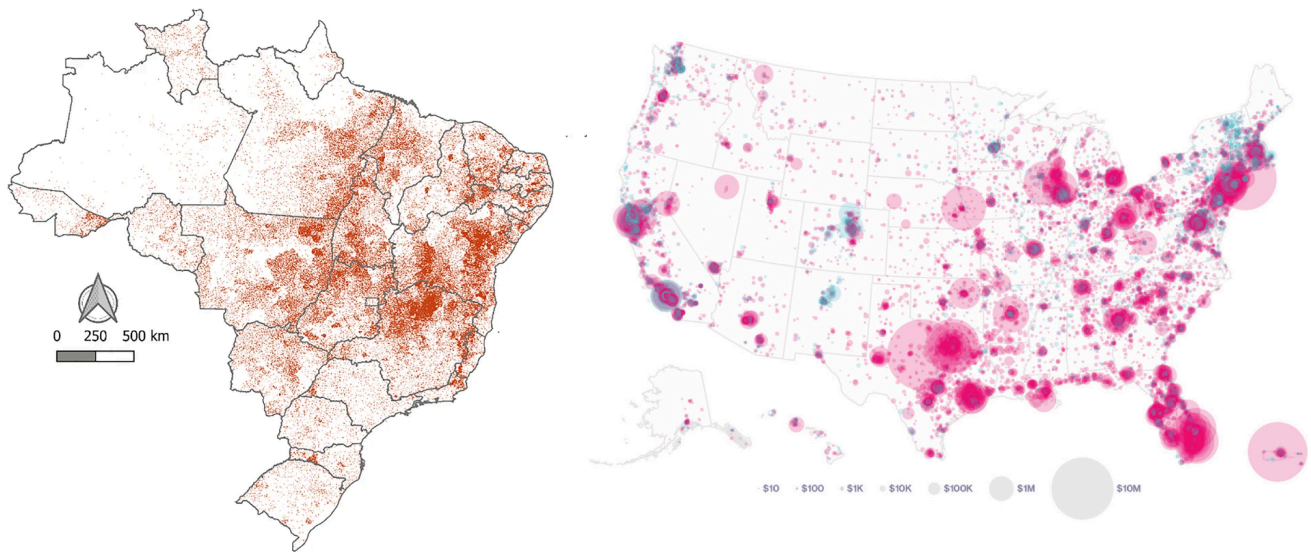
1.1 Choropleth maps



A choropleth map colours predefined areas (districts, neighbourhoods, census tracts) according to a single quantitative variable. Research shows that choropleth maps are widely used because they provide a quick overview, but they can also mislead when people interpret aggregated data as if it represents individuals. Aggregating point data into larger zones may mask spatial heterogeneity and produce a *perceptual fallacy* – users trust the map and seldom realise the effects of aggregation ([Beconytc et al., 2022](#)). To minimise these risks:

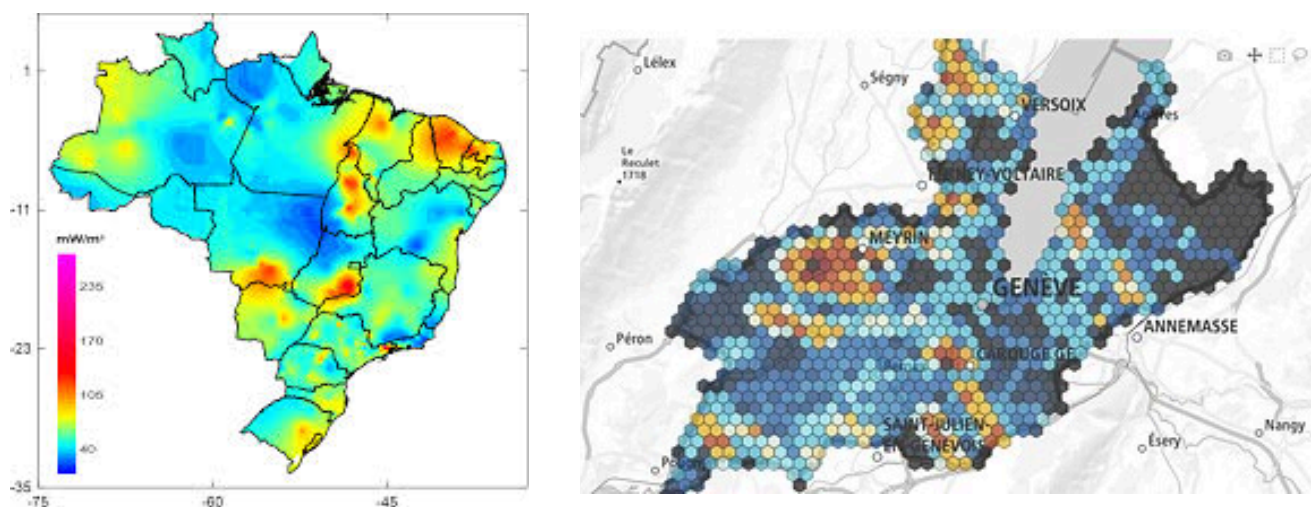
- **Normalise by population or area.** Use rates (e.g., hospital beds per 1 000 older residents) rather than raw counts. Never map counts directly, because areas with more residents will dominate the map.
- **Limit the number of classes.** Use 5–7 classes for continuous variables to avoid cognitive overload and choose classification methods carefully (e.g., quantiles versus natural breaks). The number of classes should be consistent across comparable maps.
- **Provide clear legends and units.** Legends must indicate units and class thresholds, and they should be visible at all times. A legend hidden behind a menu undermines interpretation.
- **Warn users about aggregation.** Include a note explaining that results may change with different spatial units and encourage switching units when possible (see [Section 3](#) on MAUP).

1.2 Dot density and proportional symbol maps



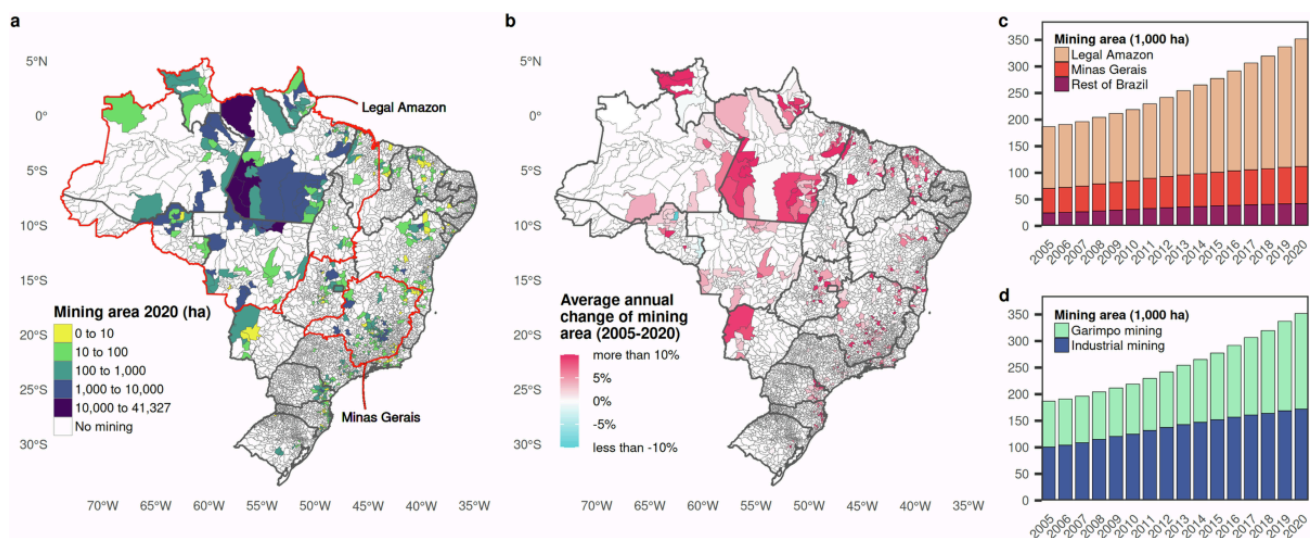
When the goal is to show counts or volumes (e.g., number of services per district), dot density maps or proportional symbols are preferable. Dot density maps randomly distribute dots within zones to represent counts; proportional symbol maps use circles or other shapes whose areas encode quantity. These methods avoid the perceptual pitfalls of colouring areas by counts and communicate magnitude more directly.

1.3 Heatmaps and hexbin plots



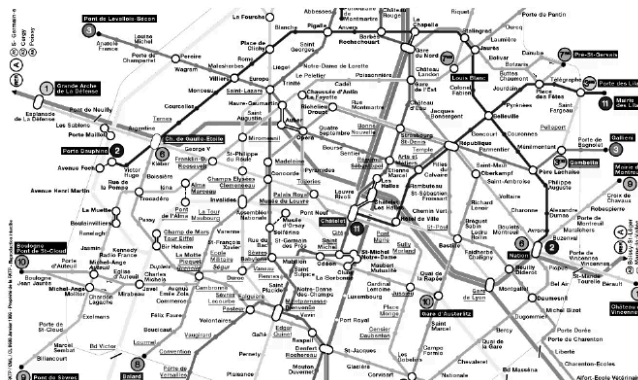
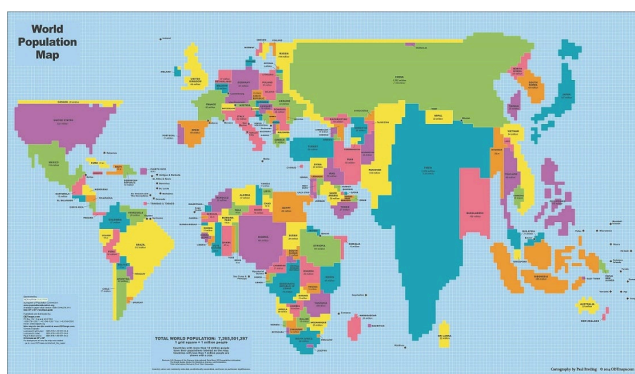
Heatmaps interpolate point data across space to depict continuous surfaces such as pollution levels or travel times. They can be appealing but require careful parameter choices (bandwidth, kernel) and may exaggerate patterns ([Beconytc et al., 2022](#)). Hexagon binning divides space into a grid of equal-sized hexagons and counts the number of points in each cell; it reduces visual clutter and provides a regular basis for analysis. Use these techniques only when the variable naturally varies continuously or when representing dense point distributions; otherwise, they may mislead.

1.4 Small multiples and linked views for multivariate analysis



The Aging Map will handle many variables (demography, services, risk factors). Attempting to encode multiple variables in a single bivariate or trivariate choropleth quickly becomes cognitively overwhelming. **Small multiples** (several small maps shown side by side) and **linked views** (map coordinated with scatterplots or bar charts) are recommended. They allow users to compare variables across space without conflating them and support exploratory analysis. Studies have shown that presenting multiple variables separately helps users formulate hypotheses and reduces misinterpretation ([Korporaal et al., 2020](#)).

1.5 Map-like alternatives: cartograms and schematic maps



Cartograms distort the sizes of areas to reflect a variable of interest (e.g., number of older residents). They can emphasise disparities that choropleth maps obscure. Schematically simplifying geography (simplifying shapes or aligning them in grids) can improve readability and facilitate infographics. A recent survey positions these techniques along a continuum from realistic maps to abstract visualisations, noting that **schematisation** reduces geographic realism to emphasise data relationships, while **imitation** uses visual metaphors to increase realism. These approaches may be useful in summary dashboards or communication materials but should supplement, not replace, the main maps.

1.6 Expanded perspectives of map-like visualization

Map-like Visualization								
Perspective	Imitation				Schematization			
	Point	Line	Area	Field	Point	Line	Area	Field
Category								
Technique	Importance labeling	Outline-centered	Grid-based	Coloring	Reposition Data Points	Border-centered	Shape Deforming	Stretching
	Map Icons	Edge-centered	Geometric Tessellation	Contouring	Reposition Nodes	Path-centered	Graphical	Density-based
			Geometric Hull				Mosaic	

Figure 1

“The two perspectives of map-like visualization: Imitation transforms visualizations to make them look like a map. Schematization transforms cartographic maps to be more visualization-like, emphasizing the display of data over geographic accuracy.”

Note. From [“The state of the art in map-like visualization.” by M. Hognräfer, M. Heitzler, and H.-J. Schulz, 2020, Computer Graphics Forum, 39\(3\), p. 647. Copyright 2020 by The Eurographics Association and John Wiley & Sons Ltd.](#)

1.7 Recommendations for the Aging Map

Visualization Technique	Recommended Use	Key Considerations
Choropleth Maps	Variables that are rates or proportions (e.g., aging rates).	Must be normalized and include clear legends and explanations.
Proportional Symbol Maps or Dot Density Maps	Raw counts (e.g., number of health centers).	Provide options to switch between counts and densities.
Small Multiples and Linked Views	Comparing multiple variables across space (e.g., population aging, service coverage).	Link adjacent maps to scatterplots for deeper comparison.
Cartograms or Schematic Maps	Summary presentations.	Detailed analysis should rely on standard maps.

2. Representing and communicating uncertainty

Spatial data often come with uncertainty due to sampling error, measurement error or modelling assumptions. A controlled experiment on map-based decision making found that visualising uncertainty influences users’ strategies rather than accuracy: participants who saw uncertainty information spent more time exploring the map, adopted more conservative decisions and displayed loss-averse behaviour. Conversely, time pressure affected how long participants looked at the legend but did not increase errors ([Korporaal et al., 2020](#)). For the Aging Map:

- **Always disclose uncertainty at an appropriate level.** For example, show margins of error for population estimates or confidence intervals for modelled indicators. Use simple visual cues such as hatching, transparency or error bars rather than complex bivariate colour schemes.
- **Offer uncertainty toggles.** Allow users to turn uncertainty layers on and off. In the default public view, show clear indicators of data quality (e.g., icons or text indicating “estimate” or “measured value”). In the research view, provide numeric measures of uncertainty on demand.
- **Explain the implications.** Include explanatory notes that caution against over-interpretation and indicate whether differences between areas are likely meaningful.

3. Modifiable areal unit problem (MAUP)

The **Modifiable Areal Unit Problem** occurs when results change depending on the size or shape of the spatial units used. Aggregating data at larger units can hide local variation, and different zoning schemes can produce opposite conclusions ([Wong, 2021](#)). This phenomenon can bias policy decisions, such as misallocating resources if high-risk pockets are averaged with low-risk areas. To mitigate MAUP:

- **Allow multiple spatial units.** Provide maps at multiple levels (e.g., census tracts, districts, subprefectures) and clearly label the unit of analysis.
- **Warn users about sensitivity.** Include documentation that demonstrates how results change when switching units and emphasise that the choice of unit can influence rankings and priorities.
- **Encourage cross-scale analysis.** Provide features that let users drill down into smaller units or aggregate up to larger ones, enabling them to detect patterns at various scales.

4. Accessibility and usability for older adults

Older adults may experience decreased vision, motor control, hearing and cognitive processing. The World Wide Web Consortium's guidance notes that designing for older users often overlaps with designing for people with disabilities ([Arch et al., 2020](#)). To ensure that the Aging Map is inclusive:

- **High contrast and large fonts.** Use high contrast colour schemes and ensure text and icons are large enough for low-vision users. Avoid relying solely on colour; use shapes or patterns to distinguish categories.
- **Large clickable targets and clear interactions.** Buttons, sliders and map markers should be large enough to be easily clicked; interactive elements must be spaced apart to prevent accidental clicks. Provide keyboard navigation for users who cannot use a mouse.
- **Plain language and clear labels.** Use simple terminology and avoid jargon. Every filter, legend and indicator should be explained in plain language or provide a tooltip with an explanation.
- **Support trial-and-error.** Research on interactive visualisations for older adults recommends designs that encourage exploration and allow easy recovery from mistakes ([Fan et al., 2023](#)). Provide undo buttons, reset views and clearly indicate changes.

5. Data governance and sustainability

The value of the Aging Map depends on the quality, transparency and longevity of its data. To achieve this:

- **Curate authoritative sources.** Use official databases (e.g., census data, health service registries) and document each dataset's source, vintage, variables, geographic coverage and licence.

- **Maintain a metadata catalogue.** For each layer, provide metadata detailing the collection methodology, update frequency and limitations. Adopt FAIR principles (findable, accessible, interoperable, reusable) to support reproducible research.
- **Plan for updates.** Set up processes for regular data updates and flag stale layers to users. Indicate the last update date prominently in the interface.
- **Protect privacy.** Use aggregated data at levels that prevent identification and comply with data protection laws. Avoid displaying sensitive indicators at very fine scales that could stigmatise small areas or reveal personal information.

6. User interface and interaction design

To serve both decision-makers and researchers, the interface should provide two modes: a **basic mode** for quick insights and a **research mode** for advanced exploration.

- **Progressive disclosure.** Present essential information first and reveal additional options (filters, variables, comparison tools) as users explore. This reduces initial complexity and supports novices while still serving experts.
- **Contextual help and storytelling.** Provide step-by-step guides or “tours” that illustrate typical tasks, such as identifying service gaps or comparing indicators. Use narrative elements (charts, callouts) to explain findings.
- **Performance optimisation.** Use vector tiles or tiling services for large datasets to ensure smooth panning and zooming. Implement data caching and pre-aggregation to avoid long load times, as delays over a few seconds reduce perceived usability ([Fan et al., 2023](#)).

7. Conclusion

This document translates the results of a comprehensive literature review into concrete design decisions for the Aging Map. By carefully selecting map types, representing

uncertainty transparently, addressing the modifiable areal unit problem, meeting accessibility guidelines for older users and planning robust data governance, the project can deliver a tool that is both scientifically rigorous and practically useful. These recommendations should guide the engineering and design phases and serve as the baseline for subsequent iterations and user studies.

AI Usage Statement

This work made limited use of generative artificial intelligence (ChatGPT v5.3) as a support tool for structuring and refining the manuscript. Specifically, the tool was used to assist in organizing sections, categorizing topics, and reviewing grammar and language clarity.

All substantive content, including the selection of references, verification of sources, interpretation of evidence, and final validation of the material, was conducted independently by the author.

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